

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$y = 2(x - d)(x + d)(x + g)(x - d)$$

In the given equation, d and g are unique positive constants. When the equation is graphed in the xy -plane, how many distinct x -intercepts does the graph have?

Answer

- A. 4
- B. 3
- C. 2
- D. 1

Correct Answer: B

Rationale

Choice B is correct. An x -intercept of a graph in the xy -plane is a point on the graph where the value of y is 0 . Substituting 0 for y in the given equation yields $0 = 2(x - d)(x + d)(x + g)(x - d)$. By the zero product property, the solutions to this equation are $x = d$, $x = -d$, $x = -g$, and $x = d$. However, $x = d$ and $x = d$ are identical. It's given that d and g are unique positive constants. It follows that the equation $0 = 2(x - d)(x + d)(x + g)(x - d)$ has 3 unique solutions: $x = d$, $x = -d$, and $x = -g$. Thus, the graph of the given equation has 3 distinct x -intercepts.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.